BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test (p=0.0133676, statistically significant at alpha=0.05). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, p=0.0133676 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.045954.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions disrupted the domain, p=0.041705 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 327 aa (permutation-corrected p=0.035964, statistically significant at alpha=0.05). This is 47 aa from the nearest configured domain boundary at 280 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=163) and "not_retained" (n=15). For Frame_status, retained: 142/152 (93.4%, 95% CI 88.3%-96.4%); not_retained: 9/11 (81.8%, 95% CI 52.3%-94.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 52.0736 vs 36.4, p=0.11965 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (67 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AGK	14	0.0782122905027933	0.13376762457977448	0.0	0.9846938775510204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2792922449674053	1
ABCC2	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	2
ITGB5	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	3
MKLN1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	4
NEO1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	5
PWWP2A	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	6
TBXAS1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	7
UBE2H	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	8
WEE2	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1549	43	0.24022346368715083	0.13376762457977448	0.0	0.436046511627907	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24559849203424552	10
CLIP1	2	0.0111731843575419	0.13376762457977448	0.0	0.7589285714285714	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22531571720546253	11
LUC7L2	1	0.00558659217877095	0.13376762457977448	0.0	0.6428571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20622902526611697	12
CUL1	5	0.027932960893854747	0.13376762457977448	0.0	0.5249999999999999	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19525436445207067	13
SND1	17	0.09497206703910614	0.13376762457977448	0.0	0.3897058823529411	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19507197864858727	14
FAM131B	3	0.01675977653631285	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19083098057337952	15
CARMIL1	2	0.0111731843575419	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19049428863403395	16
MAD1L1	2	0.0111731843575419	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19049428863403395	17
AKAP9	2	0.0111731843575419	0.13376762457977448	0.0	0.5223214285714286	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18982464577689112	18
EXOC4	2	0.0111731843575419	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18915500291974824	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ABCC1	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	20
CAPZA2	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	21
KLHL12	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	22
LRBA	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	23
OSBPL7	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	24
PRIM2	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	25
PRKAR1B	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	26
PTPRZ1	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	27
RGS3	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	28
TRA2B	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TRIM24	8	0.0446927374301676	0.13376762457977448	0.0	0.4419642857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18782694027010738	30
CCDC6	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	31
CMAS	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	32
ERC1	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	33
MINDY4	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	34
PHTF2	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	35
SNX8	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	36
AGAP3	6	0.0335195530726257	0.13376762457977448	0.0	0.4196428571428571	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1811267706771305	37
NRF1	2	0.0111731843575419	0.13376762457977448	0.0	0.4642857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18111928863403398	38
GTF2IRD1	3	0.01675977653631285	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17475955200195098	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TMPRSS2	4	0.0223463687150838	0.13376762457977448	0.0	0.37797619047619047	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17152481536986797	40
ATF7	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	41
CLEC2L	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	42
DBI	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	43
DOCK4	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	44
GIPC2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	45
GTF2I	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	46
KCND2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	47
NFIA	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	48
OSBPL9	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	49

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PAK2	1	0.0055865921787709 5	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.171407596694688 4	50
PRKAR2B	1	0.0055865921787709 5	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.171407596694688 4	51
RRBP1	1	0.0055865921787709 5	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.171407596694688 4	52
SETBP1	1	0.0055865921787709 5	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.171407596694688 4	53
SLC45A3	1	0.0055865921787709 5	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.171407596694688 4	54
SPOCK1	1	0.0055865921787709 5	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.171407596694688 4	55
ZC3HAV1	1	0.0055865921787709 5	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.171407596694688 4	56
ZNF207	1	0.0055865921787709 5	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.171407596694688 4	57
CDK5RAP2	4	0.0223463687150838	0.13376762457977448	0.0	0.3660714285714286	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.169739101084153 7	58
MKRN1	10	0.0558659217877094 94	0.13376762457977448	0.0	0.0625	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.134259252720227 1	59

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EPB41	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	60
PJA2	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	61
RBM33	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	62
RBMS3	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	63
PLPP1	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	64
SCRIB	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	65
SVOPL	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	66
TRIM33	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	67

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
238	1	162	0	15	None	1.0	-0.0
287	1	162	1	14	0.08641975308641975	0.16187392877547135	0.7908230925727642
303	2	161	1	14	0.17391304347826086	0.23330512802756184	0.6320757153442778

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
327	19	144	6	9	0.19791666666666666	0.00883450422715351	2.053817817119616
380	68	95	8	7	0.6263157894736842	0.422537724516002	0.37413451081430776
381	114	49	9	6	1.5510204081632653	0.5596655980047007	0.2520713878510084
393	140	23	12	3	1.5217391304347827	0.46313499579777323	0.33429240123968906
400	141	22	12	3	1.6022727272727273	0.44662922460047316	0.3500528628421283
438	156	7	13	2	3.4285714285714284	0.16897876327149897	0.7721678727445281
439	163	0	14	1	None	0.08426966292134831	1.0743287432532127

domain_disruption tables

frame_domain_contingency_table

142	22
9	5

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	178	14	0	164	0.0
cysteine_rich	Cysteine-rich domain	178	14	1	163	0.005699893955461294

domain_retention tables

frame_domain_contingency_table

142	21
9	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	178	163	0	15	0.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1
MAD1L1	2
MINDY4	1
MKLN1	1
MKRN1	10
NEO1	1
NFIA	1

Partner_gene	Event_count
NRF1	2
OSBPL7	1
OSBPL9	1
PAK2	1
PHTF2	1
PJA2	1
PLPP1	1
PRIM2	1
PRKAR1B	1
PRKAR2B	1
PTPRZ1	1
PWWP2A	1
RBM33	1
RBMS3	1
RGS3	1
RRBP1	1
SCRIB	1
SETBP1	1
SLC45A3	1
SND1	17
SNX8	1
SPOCK1	1
SVOPL	1
TBXAS1	1
TMPRSS2	4
TRA2B	1
TRIM24	8
TRIM33	1
UBE2H	1
WEE2	1
ZC3HAV1	1
ZNF207	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021458-T01-IM6-1	P-0021458-T01-IM6		ABCC1-BRAF	ABCC1	in-frame	three_prime	140493152	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7,16)(q34, p13.11)(chr7:g.140493152::chr16:g.16116105) | Protein Fusion: in frame {ABCC1:BRAF}	NA
EVT-P-0034621-T01-IM6-2	P-0034621-T01-IM6		ABCC2-BRAF	ABCC2	unknown	three_prime	140495500	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34; 24.2)(chr7:g.140495500::chr10:g.101559038) | Protein Fusion: mid-exon {ABCC2:BRAF}	NA
EVT-P-0052582-T01-IM6-3	P-0052582-T01-IM6		AGAP3-BRAF	AGAP3	unknown	three_prime	140485658	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+1690:BRAFinv | Protein Fusion: mid-exon {AGAP3:BRAF}	NA
EVT-P-0050574-T02-IM6-4	P-0050574-T02-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140493208	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.1140+900:BRAFinv | Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-0039799-T01-IM6-5	P-0039799-T01-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140482317	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFinv | Protein Fusion: in frame {AGAP3:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0050931-T01-IM6-6	P-0050931-T01-IM6		AGK-BRAF	AGK	in-frame	three_prime	140499251	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0042835-T03-IM6-7	P-0042835-T03-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0042835-T02-IM6-8	P-0042835-T02-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0050635-T01-IM6-9	P-0050635-T01-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140487501	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489:AKAP9_c.1141-117:BRAFinv | Protein Fusion: in frame {AKAP9:BRAF}	NA
EVT-P-0030237-T03-IM6-10	P-0030237-T03-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140492357	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751:BRAFinv | Protein Fusion: in frame {AKAP9:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023821-T01-IM6-12	P-0023821-T01-IM6		ATF7-BRAF	ATF7	in-frame	three_prime	140483233	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement : t(7;12)(q34;q13.13)(chr7:g.140483233::chr12:g. 53913299) | Protein Fusion: in frame {ATF7:BRAF}	NA
EVT-P-0007861-T01-IM5-13	P-0007861-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAFinv | Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-0021519-T02-IM6-14	P-0021519-T02-IM6		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493112	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement : c.1326+142:AGAP3_c.1140+996:BRAFinv | Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-0009757-T01-IM5-15	P-0009757-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237:BRAFinv | Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-0036886-T01-IM6-16	P-0036886-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494309	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+10859:AGK_c.981-42:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025561-T03-IM6-17	P-0025561-T03-IM6		BRAF-AGK	AGK	in-frame	three_prime	140498719	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c.980+1288:BRAF_c.101+772:AGKinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0025561-T01-IM6-18	P-0025561-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498874	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288:BRAF_c.101+772:AGKinv | Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-0024227-T01-IM6-19	P-0024227-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140495070	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796:AGK_c.981-803:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0004477-T01-IM5-20	P-0004477-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFinv | Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0023378-T01-IM6-21	P-0023378-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498362	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800:BRAF_c.101+14574:AGKinv | Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-0001832-T02-IM6-22	P-0001832-T02-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494489	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572:AGK_c.981-222:BRAFinv | Protein fusion: in frame (AGK-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000913-T02-IM5-23	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFinv | Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0029531-T01-IM6-24	P-0029531-T01-IM6		BRAF-AGK	AGK	unknown	three_prime	140500233	7	303	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAFinv | Protein Fusion: mid-exon {AGK:BRAF}	NA
EVT-P-0030746-T01-IM6-25	P-0030746-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140497533	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0010945-T01-IM5-26	P-0010945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFinv | Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0049035-T01-IM6-85	P-0049035-T01-IM6		BRAF-CLEC2L	CLEC2L	in-frame	five_prime	140485799	9	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BRAF_c.191-5932:CLEC2Linv | Protein Fusion: in frame {BRAF:CLEC2L}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004795-T01-IM5-87	P-0004795-T01-IM5		BRAF-CUL1	CUL1	in-frame	three_prime	140493811	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1(NM_003592) - BRAF (NM_004333) Rearrangement :c.790-427:CUL1_c.1140+297:BRAFinv | Protein fusion: in frame (CUL1-BRAF)	NA
EVT-P-0022257-T01-IM6-88	P-0022257-T01-IM6		BRAF-EPB41	EPB41	in-frame	three_prime	140482687	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EPB41 (NM_001166005) - BRAF (NM_004333) fusion (EPB41 exons 1-5 fused in-frame with BRAF exons 11-18): t(1;7)(p35.4;q34)(chr1:g.29424227::chr7:g.140482687) | Protein Fusion: in frame {EPB41:BRAF}	NA
EVT-P-0001988-T01-IM3-89	P-0001988-T01-IM3		BRAF-FAM131B	FAM131B	in-frame	three_prime	140489629	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BRAFdel | Protein fusion: in frame (FAM131B-BRAF)	NA
EVT-P-0010494-T01-IM5-90	P-0010494-T01-IM5		BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655) -BRAF (NM_004333) Rearrangement : t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	NA
EVT-P-0054046-T01-IM6-93	P-0054046-T01-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	140484706	10	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-1749:BRAF_c.2320+10446:GTF2IRD1inv | Protein Fusion: in frame {BRAF:GTF2IRD1}	NA
EVT-P-0000828-T03-IM6-94	P-0000828-T03-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	140483554	10	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-597:BRAF_c.2320+14181:GTF2IRD1inv | Protein Fusion: in frame {BRAF:GTF2IRD1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015847-T02-IM6-95	P-0015847-T02-IM6		BRAF-ITGB5	ITGB5	out-of-frame	five_prime	140497019	8	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - ITGB5 (NM_002213) Rearrangement : t(3,7)(q21.2,q34)(chr3:g.124579522::chr7:g.140497019) | Protein Fusion: out of frame {BRAF:ITGB5}	NA
EVT-P-0014377-T01-IM6-99	P-0014377-T01-IM6		BRAF-NFIA	NFIA	in-frame	three_prime	140486673	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NFIA (NM_001145512) - BRAF (NM_004333) rearrangement: t(1;7)(p31.3;q34)(chr1:g.61836194::chr7:g.140486673) | Protein fusion: in frame (NFIA-BRAF)	NA
EVT-P-0010398-T01-IM5-100	P-0010398-T01-IM5		BRAF-OSBPL9	OSBPL9	in-frame	three_prime	140484912	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement : t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) | Protein fusion: in frame (OSBPL9-BRAF)	NA
EVT-P-0003502-T01-IM5-101	P-0003502-T01-IM5		BRAF-PJA2	PJA2	in-frame	five_prime	140482379	10	438	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA | Protein fusion: in frame (BRAF-PJA2)	NA
EVT-P-0036870-T01-IM6-102	P-0036870-T01-IM6		BRAF-PRIM2	PRIM2	in-frame	three_prime	140493232	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRIM2 (NM_000947) - BRAF (NM_004333) Rearrangement : t(6;7)(p11.1;q34)(chr6:g.57410815::chr7:g.140493232) | Protein Fusion: in frame {PRIM2:BRAF}	NA
EVT-P-0009246-T01-IM5-103	P-0009246-T01-IM5		BRAF-RBM33	RBM33	in-frame	three_prime	140482661	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526:RBM33_c.1314+160:BRAFinv | Protein fusion: in frame (RBM33-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016313-T01-IM6-104	P-0016313-T01-IM6		BRAF-SLC45A3	SLC45A3	unknown	three_prime	140486608	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SLC45A3 (NM_033102) - BRAF (NM_004333) fusion (SLC45A3 exons 1 fused to BRAF exons 10-18): t(1;7)(q32.1;q34)(chr1:g.205638452::chr7:g.140486608) | Transcript Fusion {SLC45A3:BRAF}	NA
EVT-P-0064764-T01-IM7-105	P-0064764-T01-IM7		BRAF-SND1	SND1	in-frame	five_prime	140493289	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - SND1 (NM_014390) fusion: c.1140+819:BRAF_c.1039-3950:SND1inv | Protein Fusion: in frame {BRAF:SND1}	NA
EVT-P-0029981-T01-IM6-106	P-0029981-T01-IM6		BRAF-SPOCK1	SPOCK1	in-frame	three_prime	140483862	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SPOCK1 (NM_004598) - BRAF (NM_004333) fusion: t(5;7)(q31.2;q34)(chr5:g.136401840::chr7:g.140483862) | Protein Fusion: in frame {SPOCK1:BRAF}	NA
EVT-P-0064223-T01-IM7-107	P-0064223-T01-IM7		BRAF-TBXAS1	TBXAS1	out-of-frame	five_prime	140494512	8	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - TBXAS1 (NM_001061) rearrangement: c.981-245:BRAF_c.336+2985:TBXAS1inv | Protein Fusion: out of frame {BRAF:TBXAS1}	NA
EVT-P-0021469-T01-IM6-108	P-0021469-T01-IM6		BRAF-TRIM33	TRIM33	in-frame	three_prime	140481644	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM33 (NM_015906) - BRAF(NM_004333) rearrangement: t(1;7)(p13.2;q34)(chr1:g.114963985::chr7:g.140481644) | Protein Fusion: in frame {TRIM33:BRAF}	NA
EVT-P-0042835-T03-IM6-109	P-0042835-T03-IM6		BRAF-WEE2	WEE2	unknown	five_prime	140498912	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - WEE2 (NM_001105558) rearrangement: c.980+1250:BRAF_c.1305:WEE2inv | Protein Fusion: mid-exon {BRAF:WEE2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015985-T03-IM6-111	P-0015985-T03-IM6		CAPZA2-BRAF	CAPZA2	in-frame	three_prime	140491425	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CAPZA2 (NM_006136) - BRAF (NM_004333) fusion: c.40-938:CAPZA2_c.1140+2683:BRAFinv | Protein Fusion: in frame {CAPZA2:BRAF}	NA
EVT-P-0033764-T04-IM6-112	P-0033764-T04-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	140491509	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion: t(6;7)(p22.2;q34)(chr6:g.25554753::chr7:g.140491509) | Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-0030796-T01-IM6-113	P-0030796-T01-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	140491645	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion (LRRC16A exons 1-29 fused to BRAF exons 9-18): t(6;7)(p22.2;q34)(chr6:g.25580104::chr7:g.140491645) | Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-0005834-T01-IM5-114	P-0005834-T01-IM5		CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CCDC6 (NM_005436) -BRAF (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) | Protein fusion: in frame (CCDC6-BRAF)	NA
EVT-P-0038194-T01-IM6-115	P-0038194-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140484954	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(q34;q33.2)(chr7:g.140484954::chr9:g.123187156) | Protein Fusion: in frame {CDK5RAP2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004203-T01-IM5-116	P-0004203-T01-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140491585	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(7q34; 9q33.2)(chr7:140491585::chr9:123251876) | Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-0005461-T02-IM5-117	P-0005461-T02-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140481684	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement : t(7;9)(7q34; 9q33.2)(chr7:140481684::chr9:123255452) | Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-0014969-T01-IM6-118	P-0014969-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	five_prime	140493700	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CDK5RAP2 (NM_018249) Rearrangement : t(7;9)(q34; q33.3)(chr7:g.140493700::chr9:g.123269531) | Protein fusion: in frame (BRAF-CDK5RAP2)	NA
EVT-P-0040461-T01-IM6-119	P-0040461-T01-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140488218	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140488218::chr12:g.122833264) | Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-0007266-T02-IM6-120	P-0007266-T02-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140495673	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140495673::chr12:g.122767026) | Protein Fusion: in frame {CLIP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024995-T01-IM6-121	P-0024995-T01-IM6		CMAS-BRAF	CMAS	unknown	three_prime	140488046	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.22218755) | Transcript Fusion {CMAS:BRAF}	NA
EVT-P-0039648-T01-IM6-122	P-0039648-T01-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	140492893	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAFinv | Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0039648-T03-IM6-123	P-0039648-T03-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	140492893	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAFinv | Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0064088-T01-IM7-124	P-0064088-T01-IM7		CUL1-BRAF	CUL1	in-frame	three_prime	140489451	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+1956:CUL1_c.1141-2067:BRAFinv | Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0068173-T01-IM7-125	P-0068173-T01-IM7		CUL1-BRAF	CUL1	in-frame	three_prime	140491990	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+191:CUL1_c.1140+2118:BRAFinv | Protein Fusion: in frame {CUL1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0050196-T01-IM6-126	P-0050196-T01-IM6		DBI-BRAF	DBI	in-frame	three_prime	140486284	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(chr2:g.120128894::chr7:g.140486284) | Protein Fusion: in frame {DBI:BRAF}	NA
EVT-P-0064588-T01-IM7-127	P-0064588-T01-IM7		DOCK4-BRAF	DOCK4	out-of-frame	three_prime	140483872	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298:DOCK4_c.1178-915:BRAFdup | Protein Fusion: out of frame {DOCK4:BRAF}	NA
EVT-P-0014833-T02-IM6-128	P-0014833-T02-IM6		ERC1-BRAF	ERC1	in-frame	three_prime	140490668	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1246645) | Protein Fusion: in frame {ERC1:BRAF}	NA
EVT-P-0036937-T01-IM6-129	P-0036937-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140490241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857:BRAFinv | Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-0024206-T01-IM6-130	P-0024206-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140487956	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) rearrangement: c.87-8072:EXOC4_c.1141-572:BRAFinv | Protein Fusion: out of frame {EXOC4:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047918-T04-IM7-131	P-0047918-T04-IM7		FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel | Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-0047918-T02-IM6-132	P-0047918-T02-IM6		FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel | Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-0044845-T01-IM6-134	P-0044845-T01-IM6		GTF2I-BRAF	GTF2I	unknown	three_prime	140485631	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2I (NM_032999) - BRAF (NM_004333) fusion: c.768:GTF2I_c.1177+1717:BRAFinv | Protein Fusion: mid-exon {GTF2I:BRAF}	NA
EVT-P-0000828-T05-IM6-135	P-0000828-T05-IM6		GTF2IRD1-BRAF	GTF2IRD1	in-frame	three_prime	140483546	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2IRD1 (NM_005685) - BRAF (NM_004333) rearrangement: c.2320+14147:GTF2IRD1_c.1178-589:BRAFinv | Protein Fusion: in frame {GTF2IRD1:BRAF}	NA
EVT-P-0046745-T01-IM6-136	P-0046745-T01-IM6		KCND2-BRAF	KCND2	out-of-frame	three_prime	140485867	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.1177+1481:BRAFinv | Protein Fusion: out of frame {KCND2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0056952-T01-IM6-138	P-0056952-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493821	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0044270-T02-IM6-139	P-0044270-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491645	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1549_c.1140+2463:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0027592-T01-IM6-140	P-0027592-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492343	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0055099-T03-IM6-141	P-0055099-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490663	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1549_c.1141-3279:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0058584-T01-IM7-142	P-0058584-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491493	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1663:KIAA1549_c.1140+2615:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0066030-T01-IM7-143	P-0066030-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482785	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035153-T01-IM6-144	P-0035153-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035153-T03-IM6-145	P-0035153-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140+3015:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0049925-T01-IM6-146	P-0049925-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489497	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035844-T03-IM7-147	P-0035844-T03-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035844-T04-IM7-148	P-0035844-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T02-IM6-149	P-0017693-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-495:KIAA1549_c.1141-2153:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0062963-T01-IM7-150	P-0062963-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482198	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0012575-T01-IM5-151	P-0012575-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490295	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0052411-T05-IM7-152	P-0052411-T05-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034117-T01-IM6-153	P-0034117-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140483812	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0052411-T03-IM6-154	P-0052411-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0009097-T01-IM5-155	P-0009097-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0005121-T02-IM5-156	P-0005121-T02-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491248	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0064872-T01-IM7-157	P-0064872-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489277	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013777-T01-IM5-158	P-0013777-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481674	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0040760-T01-IM6-159	P-0040760-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481540	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0014191-T01-IM6-160	P-0014191-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492679	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0014381-T01-IM6-161	P-0014381-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492888	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0028611-T01-IM6-162	P-0028611-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015540-T01-IM6-163	P-0015540-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489119	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0062070-T01-IM7-164	P-0062070-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490443	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0022404-T01-IM6-165	P-0022404-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482177	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-2270:KIAA1549_c.1314+644:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0004032-T01-IM5-166	P-0004032-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487892	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0043372-T01-IM6-167	P-0043372-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487523	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2204:KIAA1549_c.1141-139:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028611-T02-IM6-168	P-0028611-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0043874-T01-IM6-169	P-0043874-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490001	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0069024-T04-IM7-170	P-0069024-T04-IM7		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492117	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0029361-T01-IM6-171	P-0029361-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489625	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0016557-T01-IM6-172	P-0016557-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488576	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032982-T01-IM6-173	P-0032982-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491506	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0064310-T04-IM7-174	P-0064310-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493446	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0021127-T01-IM6-175	P-0021127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491741	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T01-IM6-176	P-0017693-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495:KIAA1549_c.1141-2153:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0020125-T01-IM6-177	P-0020125-T01-IM6		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492175	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045683-T02-IM6-178	P-0045683-T02-IM6		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492850	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0019127-T01-IM6-179	P-0019127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490657	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066022-T01-IM7-180	P-0066022-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488035	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0036680-T01-IM6-181	P-0036680-T01-IM6		KLHL12-BRAF	KLHL12	in-frame	three_prime	140493984	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32.1;q34)(chr1:g.202880970::chr7:g.140493984) | Protein Fusion: in frame {KLHL12:BRAF}	NA
EVT-P-0047833-T02-IM6-183	P-0047833-T02-IM6		LRBA-BRAF	LRBA	in-frame	three_prime	140493444	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31.3;q34)(chr4:g.151502269::chr7:g.140493444) | Protein Fusion: in frame {LRBA:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006412-T01-IM5-184	P-0006412-T01-IM5		LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRAFdel | Antisense fusion	NA
EVT-P-0026158-T04-IM6-185	P-0026158-T04-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359:BRAFdup | Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-0026158-T01-IM6-186	P-0026158-T01-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-9863_c.1140+2359dup | Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-0006601-T03-IM6-187	P-0006601-T03-IM6		MINDY4-BRAF	MINDY4	in-frame	three_prime	140490696	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM188B (NM_032222) - BRAF(NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312:BRAF | Protein Fusion: in frame {FAM188B:BRAF}	NA
EVT-P-0023220-T01-IM6-188	P-0023220-T01-IM6		MKLN1-BRAF	MKLN1	in-frame	three_prime	140496796	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070:MKLN1_c.981-2529:BRAFinv | Protein Fusion: in frame {MKLN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035393-T02-IM6-189	P-0035393-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140488436	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052:BRAFdup | Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0009024-T01-IM5-190	P-0009024-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623:BRAFdup | Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0008083-T03-IM6-191	P-0008083-T03-IM6		MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484511	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432:MKRN1_c.1178-1554:BRAFdup | Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-0001036-T01-IM3-192	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0047135-T01-IM6-193	P-0047135-T01-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140492510	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598:BRAFdup | Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004373-T02-IM5-194	P-0004373-T02-IM5		MKRN1-BRAF	MKRN1	unknown	three_prime	140434466	18	744	False	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup | Protein fusion: mid-exon (MKRN1-BRAF)	NA
EVT-P-0009024-T02-IM6-195	P-0009024-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-185:MKRN1_c.1140+623:BRAFdup | Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0031896-T02-IM6-196	P-0031896-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482581	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-4 fused to BRAF exons 11-18): c.772-1018:MKRN1_c.1314+240:BRAFdup | Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0008083-T02-IM6-197	P-0008083-T02-IM6		MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484505	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-2 fused with BRAF exons 10-18): c.315-428:MKRN1_c.1178-1548:BRAFdup | Protein Fusion: out of frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013536-T01-IM5-198	P-0013536-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	three_prime	140483618	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.544+172: MKRN1_c.1178-661:BRAFdup | Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0048036-T01-IM6-199	P-0048036-T01-IM6		NEO1-BRAF	NEO1	in-frame	three_prime	140497651	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NEO1 (NM_002499) - BRAF (NM_004333) fusion: t(7;15)(q34;q24.1)(chr7:g.140497651::chr15:g.73575682) | Protein Fusion: in frame {NEO1:BRAF}	NA
EVT-P-0043972-T03-IM6-200	P-0043972-T03-IM6		NRF1-BRAF	NRF1	in-frame	three_prime	140487712	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.607-7410:NRF1_c.1141-328:BRAFinv | Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-0065446-T01-IM7-201	P-0065446-T01-IM7		NRF1-BRAF	NRF1	in-frame	three_prime	140483810	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.1349-10893:NRF1_c.1178-853:BRAFinv | Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-0059002-T01-IM7-202	P-0059002-T01-IM7		OSBPL7-BRAF	OSBPL7	in-frame	three_prime	140491530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL7 (NM_145798) - BRAF (NM_004333) fusion: t(7;17)(q34;q21.32)(chr7:g.140491530::chr17:g.45890381) | Protein Fusion: in frame {OSBPL7:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054958-T02-IM6-203	P-0054958-T02-IM6		PAK2-BRAF	PAK2	in-frame	three_prime	140486065	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PAK2 (NM_002577) - BRAF (NM_004333) rearrangement: t(3;7)(q29;q34)(chr3:g.196530579::chr7:g.140486065) | Protein Fusion: in frame {PAK2:BRAF}	NA
EVT-P-0008789-T01-IM5-205	P-0008789-T01-IM5		PHTF2-BRAF	PHTF2	in-frame	three_prime	140489236	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.2224-367:PHTF2_c.1141-1852:BRAFinv | Protein fusion: in frame (PHTF2-BRAF)	NA
EVT-P-0044819-T01-IM6-206	P-0044819-T01-IM6		PLPP1-BRAF	PLPP1	out-of-frame	three_prime	140482123	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PPAP2A (NM_176895) - BRAF (NM_004333) fusion: t(5;7)(q11.2;q34)(chr5:g.54789192::chr7:g.140482123) | Protein Fusion: out of frame {PPAP2A:BRAF}	NA
EVT-P-0009694-T01-IM5-207	P-0009694-T01-IM5		PRKAR1B-BRAF	PRKAR1B	in-frame	three_prime	140491855	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR1B (NM_001164758) -BRAF (NM_004333) Rearrangement : c.892-2551:PRKAR1B_c.1140+2253:BRAFdup | Protein fusion: in frame (PRKAR1B-BRAF)	NA
EVT-P-0008567-T01-IM5-208	P-0008567-T01-IM5		PRKAR2B-BRAF	PRKAR2B	in-frame	three_prime	140486416	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063:PRKAR2B_c.1177+932:BRAFinv | Protein fusion: in frame (PRKAR2B-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041181-T01-IM6-209	P-0041181-T01-IM6		PTPRZ1-BRAF	PTPRZ1	in-frame	three_prime	140492428	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457:PTPRZ1_c.1140+1680:BRAFinv | Protein Fusion: in frame {PTPRZ1:BRAF}	NA
EVT-P-0012884-T01-IM5-210	P-0012884-T01-IM5		PWWP2A-BRAF	PWWP2A	in-frame	three_prime	140495243	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PWWP2A (NM_001130864) - BRAF(NM_004333) Rearrangement : t(5;7)(q33.3;q34)(chr5:g.159539324::chr7:g.140495243) | Protein fusion: in frame (PWWP2A-BRAF)	NA
EVT-P-0068260-T01-IM7-211	P-0068260-T01-IM7		RBMS3-BRAF	RBMS3	in-frame	three_prime	140482707	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140482707) | Protein Fusion: in frame {RBMS3:BRAF}	NA
EVT-P-0064764-T01-IM7-212	P-0064764-T01-IM7		RGS3-BRAF	RGS3	out-of-frame	three_prime	140493282	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RGS3 (NM_144488) - BRAF (NM_004333) fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.116291333) | Protein Fusion: out of frame {RGS3:BRAF}	NA
EVT-P-0041308-T02-IM6-213	P-0041308-T02-IM6		RRBP1-BRAF	RRBP1	in-frame	three_prime	140485917	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr20:g.17634546) | Protein Fusion: in frame {RRBP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007392-T01-IM5-214	P-0007392-T01-IM5		SCRIB-BRAF	SCRIB	in-frame	three_prime	140482150	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675) | Protein fusion: in frame (SCRIB-BRAF)	NA
EVT-P-0056639-T01-IM6-215	P-0056639-T01-IM6		SETBP1-BRAF	SETBP1	in-frame	three_prime	140486594	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SETBP1 (NM_015559) - BRAF (NM_004333) fusion: t(7;18)(q34;q12.3)(chr7:g.140486594::chr18:g.42638299) | Protein Fusion: in frame {SETBP1:BRAF}	NA
EVT-P-0048950-T01-IM6-216	P-0048950-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140492033	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+30037:SND1_c.1140+2075:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0012395-T01-IM5-217	P-0012395-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140493527	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0009619-T01-IM5-218	P-0009619-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482494	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052299-T04-IM6-219	P-0052299-T04-IM6		SND1-BRAF	SND1	unknown	three_prime	140482935	10	400	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1038+2929:SND1_c.1200:BRAFinv | Protein Fusion: mid-exon {SND1:BRAF}	NA
EVT-P-0001687-T01-IM3-220	P-0001687-T01-IM3		SND1-BRAF	SND1	in-frame	three_prime	140490224	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0004103-T01-IM5-221	P-0004103-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140481685	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799:SND1_c.1315-192:BRAFinv. | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0029727-T01-IM6-222	P-0029727-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140482257	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095:SND1_c.1314+564:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0068194-T01-IM7-223	P-0068194-T01-IM7		SND1-BRAF	SND1	in-frame	three_prime	140489175	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008140-T01-IM5-224	P-0008140-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489439	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0034168-T01-IM6-225	P-0034168-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489421	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18): c.1152+6972:SND1_c.1141-2037:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0025075-T01-IM6-226	P-0025075-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140490546	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1- 9 with BRAF exons 9-18) : c.1039-2471:SND1_c.1141-3162:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0002071-T02-IM5-227	P-0002071-T02-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489292	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SND1_c.1141-1908:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0002197-T01-IM3-228	P-0002197-T01-IM3		SND1-BRAF	SND1	in-frame	five_prime	140488303	9	381	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		c.1528-8436:SND1_c.1141-922:BRAFinv | Protein fusion: in frame (BRAF-SND1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007533-T01-IM5-229	P-0007533-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482226	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SND1_c.1314+595:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0036567-T01-IM6-230	P-0036567-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140493552	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SND1_c.1140+556:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0034845-T01-IM6-231	P-0034845-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SND1_c.1141-1857:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0054364-T01-IM6-232	P-0054364-T01-IM6		SNX8-BRAF	SNX8	in-frame	three_prime	140488730	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SNX8 (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNX8_c.1141-1346:BRAFdup | Protein Fusion: in frame {SNX8:BRAF}	NA
EVT-P-0026773-T01-IM6-234	P-0026773-T01-IM6		SVOPL-BRAF	SVOPL	unknown	five_prime	140482097	11	439	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) rearrangement: c.1315-604:BRAF_chr7:g.138277426del | Transcript Fusion {BRAF:SVOPL}	NA
EVT-P-0051214-T01-IM6-235	P-0051214-T01-IM6		TMPRSS2-BRAF	TMPRSS2	in-frame	three_prime	140502592	6	238	True	retained	1.0	False	Protein kinase domain	RAS-binding domain	Cysteine-rich domain	TMPRSS2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.140502592::chr21:g.42867846) | Protein Fusion: in frame {TMPRSS2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018524-T01-IM6-236	P-0018524-T01-IM6		TMPRSS2-BRAF	TMPRSS2	out-of-frame	three_prime	140490337	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(21;7)(q22.3;q34)(chr21:g.42873801::chr7:g.140490337) | Protein Fusion: out of frame {TMPRSS2:BRAF}	NA
EVT-P-0022778-T01-IM6-237	P-0022778-T01-IM6		TMPRSS2-BRAF	TMPRSS2	in-frame	three_prime	140483147	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(7;21)(q34;q22.3)(chr7:g.140483147::chr21:g.42878796) | Protein Fusion: in frame {TMPRSS2:BRAF}	NA
EVT-P-0043836-T01-IM6-239	P-0043836-T01-IM6		TRA2B-BRAF	TRA2B	in-frame	three_prime	140491978	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRA2B (NM_004593) - BRAF (NM_004333) fusion: t(3;7)(q27.2;q34)(chr3:g.185650761::chr7:g.140491978) | Protein Fusion: in frame {TRA2B:BRAF}	NA
EVT-P-0054430-T02-IM6-240	P-0054430-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487875	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2719-489:TRIM24_c.1141-491:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0025699-T02-IM6-241	P-0025699-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140486854	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2014+674:TRIM24_c.1177+494:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA

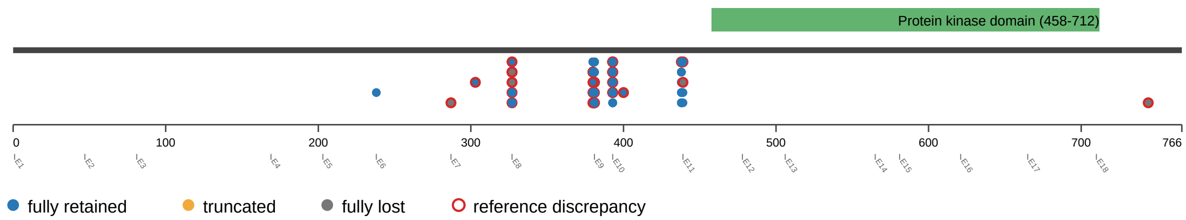
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045111-T01-IM6-242	P-0045111-T01-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140488049	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2257-760:TRIM24_c.1141-665:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0066835-T01-IM7-243	P-0066835-T01-IM7		TRIM24-BRAF	TRIM24	in-frame	three_prime	140490589	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1879-502:TRIM24_c.1141-3205:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0068451-T01-IM7-244	P-0068451-T01-IM7		TRIM24-BRAF	TRIM24	in-frame	three_prime	140482765	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.484-860:TRIM24_c.1314+56:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0030988-T01-IM6-245	P-0030988-T01-IM6		TRIM24-BRAF	TRIM24	unknown	three_prime	140491002	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2349:TRIM24_c.1140+3106:BRAFinv | Protein Fusion: mid-exon {TRIM24:BRAF}	NA
EVT-P-0032569-T01-IM6-246	P-0032569-T01-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487936	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) Rearrangement : c.1878+791:TRIM24_c.1141-552inv | Protein Fusion: in frame {TRIM24:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032569-T02-IM6-247	P-0032569-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487936	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1878+791:TRIM24_c.1141-552:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0019039-T01-IM6-248	P-0019039-T01-IM6		UBE2H-BRAF	UBE2H	out-of-frame	three_prime	140499248	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - UBE2H (NM_003344) rearrangement: c.980+914:BRAF_c.299-4611:UBE2Hdup | Protein Fusion: out of frame {UBE2H:BRAF}	NA
EVT-P-0027762-T01-IM6-250	P-0027762-T01-IM6		ZC3HAV1-BRAF	ZC3HAV1	in-frame	three_prime	140485946	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup | Protein Fusion: in frame {ZC3HAV1:BRAF}	NA
EVT-P-0008920-T01-IM5-251	P-0008920-T01-IM5		ZNF207-BRAF	ZNF207	in-frame	three_prime	140485847	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.30687533) | Protein fusion: in frame (ZNF207-BRAF)	NA

Figures

domain retention outliers

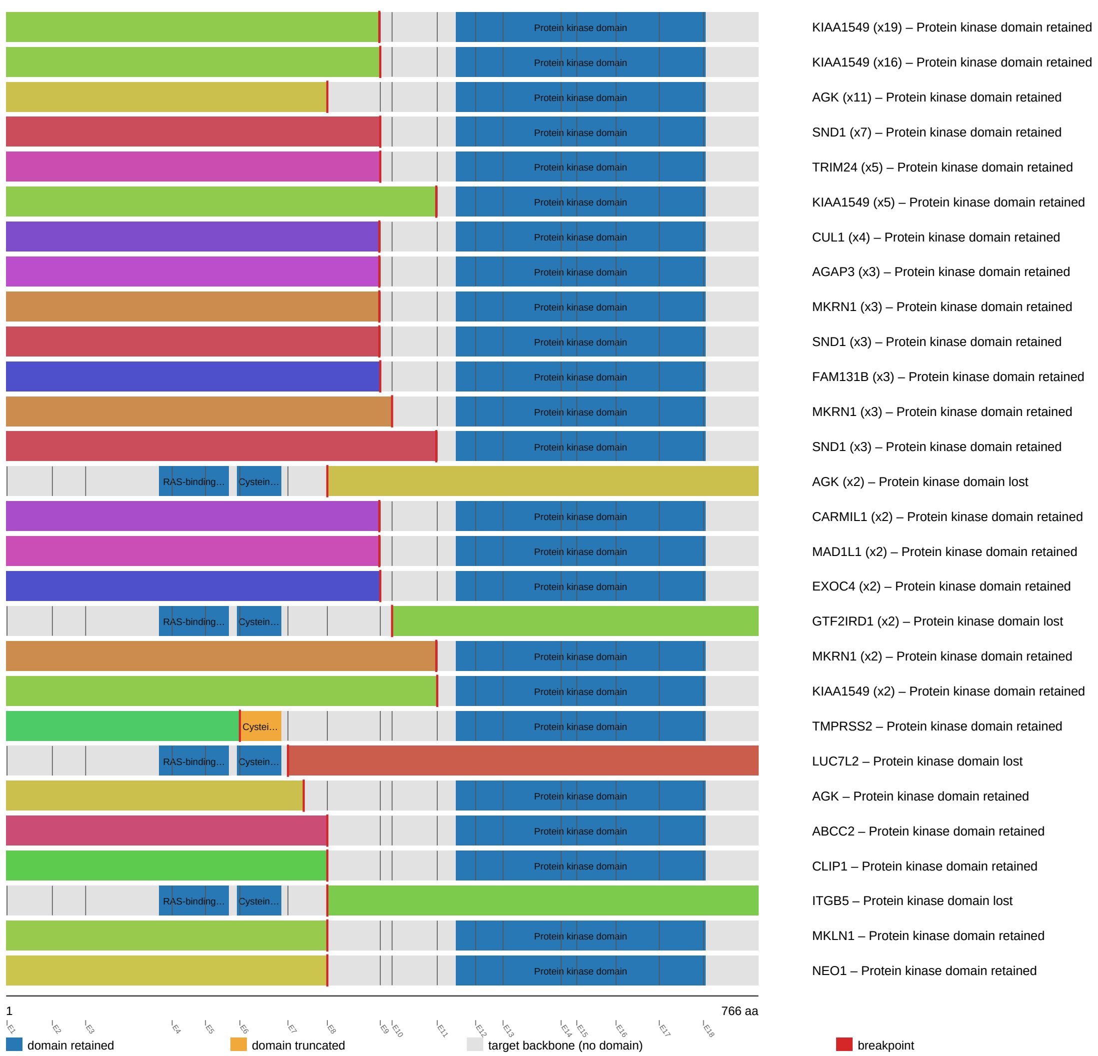
BRAF domain-retention track



fusion schematic

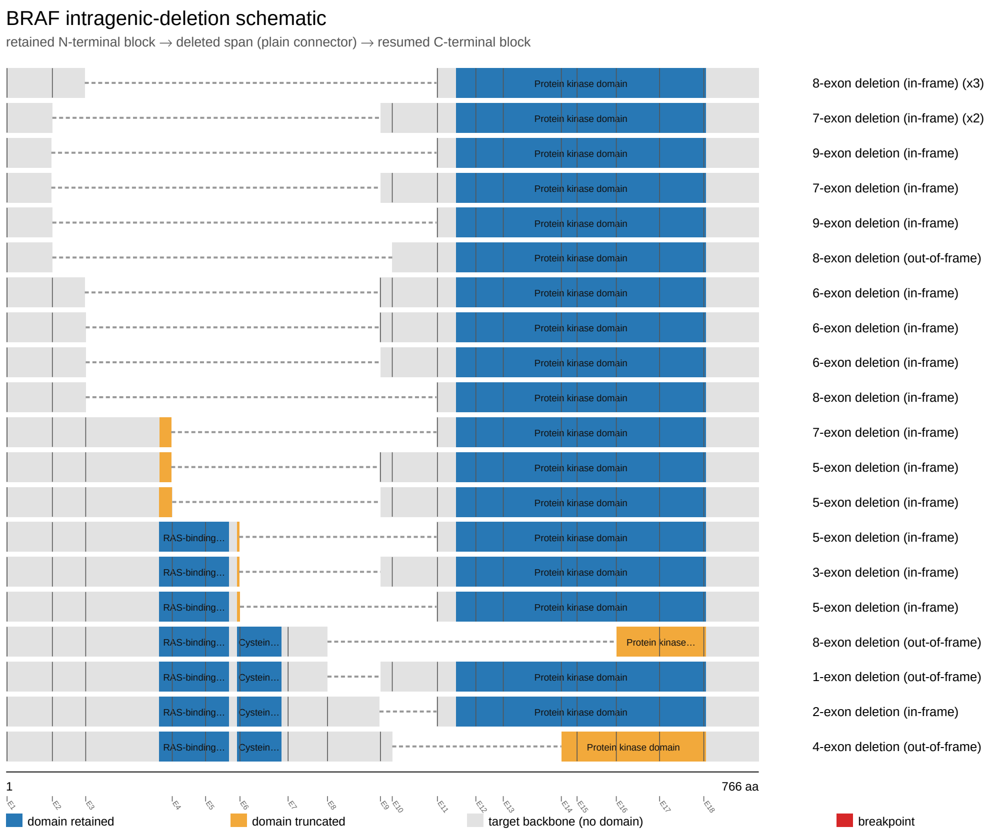
BRAF fusion-transcript schematic

partner block → breakpoint → retained BRAF portion (domain-colored, exon ticks)



Showing the top 28 of 99 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison